Hawaiian Market Affordability Analysis: Solo Travelers vs. Families

Executive Summary

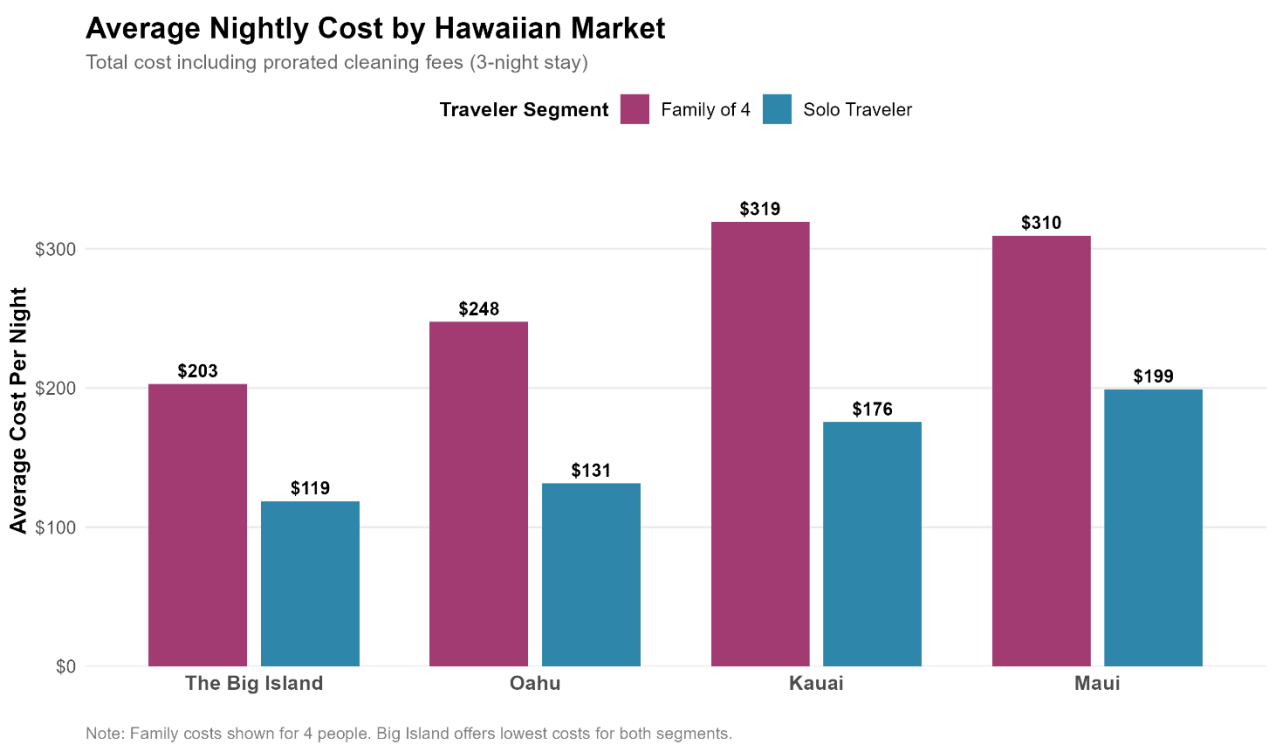
This analysis evaluates affordability across four Hawaiian Airbnb markets; The Big Island, Oahu, Kauai, and Maui, for solo budget travelers and families of four or more respectively. **Key Finding: The Big Island delivers unmatched value for both segments, providing solo travelers with twice the budget-friendly inventory of any competitor, and families with per-person costs 35-57% lower than Maui, Kauai, and Oahu.**

Metrics, Criteria, and Analytic Process

We measured affordability using total nightly cost (including prorated cleaning fees over a 3-night stay) for solo travelers in units accommodating 1-2 guests, and per-person nightly cost for families in properties with 2+ bedrooms accommodating 4+ people. For solo travelers, we categorized inventory into price tiers (ultra-budget <$75, budget $75-100, moderate $101-150, premium >$150) to assess availability of affordable options, while family analysis included evaluation of essential amenities such as kitchens, washers, and free parking.

Key Insights and Findings

- **Dramatic price variance across markets**: Solo traveler costs range from $119/night (Big Island) to $199/night (Maui), a 67% difference, while family per-person costs vary from $51 to $80, representing up to a 57% premium for the most expensive islands.
- **Family amenities are consistently excellent across all markets**: 95-100% of family-sized properties include kitchens and washers across all four islands, indicating that affordability differences are not driven by amenity gaps.
- **Premium positioning opportunity for Maui and Kauai**: These markets command 40-70% price premiums over The Big Island, suggesting they should continue to be marketed as luxury/exclusive destinations rather than competing on affordability metrics.



The Rise of the Three-Pointer: Offensive Evolution in the NBA (1990s vs. 2000s)

Executive Summary

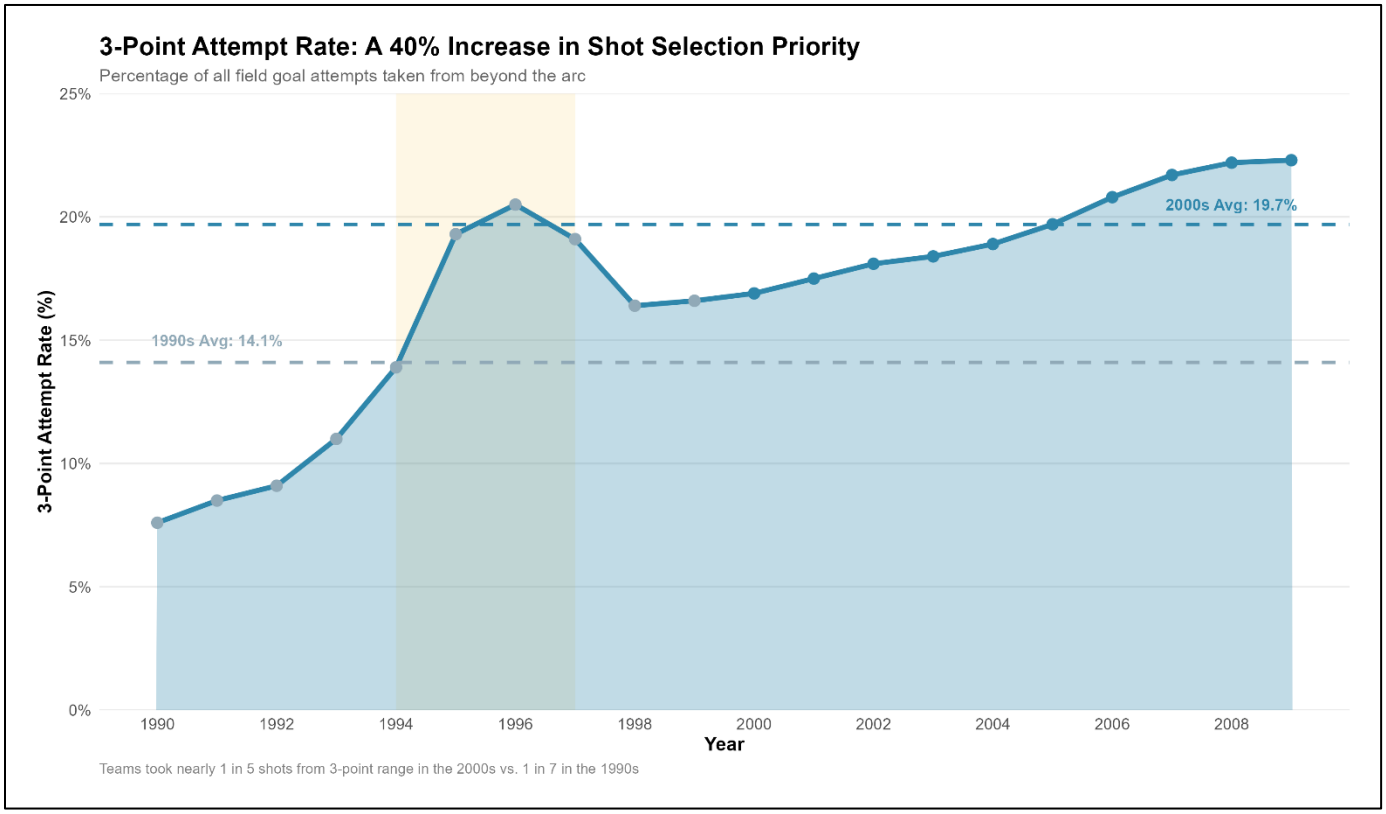
This analysis quantifies how the strategic importance of 3-point field goals changed between the 1990s and 2000s by measuring five key metrics: 3-point attempt rate (percentage of all shots taken from beyond the arc), scoring contribution (percentage of total points from threes), volume (average made per game), attempts (average shots taken), and efficiency (shooting percentage). **Key Insight: The 3-point shot evolved from a peripheral offensive weapon in the 1990s to a more central offensive strategic element in the 2000s, with teams nearly doubling their reliance on the three-pointer as both an attempt strategy (from 14% to 20% of all shots) and scoring source (from 12% to 17% of points), representing a fundamental transformation in NBA offensive philosophy.**

Metrics, Criteria, and Analytic Process

To quantify the "relative importance" of 3-point field goals, we measured five key metrics: (1) 3-point attempt rate (percentage of all field goal attempts taken from beyond the arc), (2) scoring contribution (percentage of total points from 3-pointers), (3) volume (average 3-pointers made per team per game), (4) attempts (average 3-point shots attempted per game), and (5) efficiency (3-point shooting percentage). These metrics collectively reveal how teams integrated the three-pointer into their offensive systems.

Key Insights Findings

- The NBA shortened the 3-point line from 1994-1997, causing a dramatic spike (highlighted in pale yellow in the below visual): teams made 5.96 threes per game in 1996 (nearly as many as the 2000s average)
- Teams took 14.1% of their shots from 3-point range in the 1990s versus 19.7% in the 2000s, a near-40% increase in 3pt shot selection prioritization
- Shooting percentage only marginally improved from 34.8% (1990s) to 35.6% (2000s), indicating teams were not getting much better at shooting threes, just shooting a higher volume



Appendix

Question 1:

```
db = db.getSiblingDB("sample_airbnb")
db.listingsAndReviews.aggregate([
  {
    $match: {
      "address.market": { $in: ["Maui", "Kauai", "The Big Island", "Oahu"] },
      accommodates: { $lte: 2 }
    }
  },
  {
    $project: {
      market: "$address.market",
      price: { $toDouble: "$price" },
      cleaning_fee: {
        $cond: [
          { $ifNull: ["$cleaning_fee", false] },
          { $toDouble: "$cleaning_fee" },
          0
        ]
      },
      per_night_total: {
        $divide: [
          {
            $add: [
              { $multiply: [{ $toDouble: "$price" }, 3] },
              {
                $cond: [
                  { $ifNull: ["$cleaning_fee", false] },
                  { $toDouble: "$cleaning_fee" },
                  0
                ]
              }
            ]
          },
          3
        ]
      }
    }
  },
  {
```

The Rise of the Three-Pointer: Offensive Evolution in the NBA (1990s vs. 2000s)

```
$group: {
  _id: "$market",
  avg_nightly_price: { $avg: "$price" },
  avg_cleaning_fee: { $avg: "$cleaning_fee" },
  avg_per_night_total: { $avg: "$per_night_total" },
  listing_count: { $sum: 1 },
  ultra_budget_count: {
    $sum: {
      $cond: [{ $lt: ["$per_night_total", 75] }, 1, 0]
    }
  },
  budget_count: {
    $sum: {
      $cond: [
        {
          $and: [
            { $gte: ["$per_night_total", 75] },
            { $lte: ["$per_night_total", 100] }
          ]
        },
        1,
        0
      ]
    }
  },
  moderate_count: {
    $sum: {
      $cond: [
        {
          $and: [
            { $gt: ["$per_night_total", 100] },
            { $lte: ["$per_night_total", 150] }
          ]
        },
        1,
        0
      ]
    }
  },
```

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```
        premium_count: {
            $sum: {
                $cond: [{ $gt: ["$per_night_total", 150] }, 1, 0]
            }
        }
    }
},
{
    $project: {
        _id: 0,
        market: "$_id",
        avg_nightly_price: { $round: ["$avg_nightly_price", 2] },
        avg_cleaning_fee: { $round: ["$avg_cleaning_fee", 2] },
        avg_total_per_night: { $round: ["$avg_per_night_total", 2] },
        total_listings: "$listing_count",
        pct_ultra_budget: {
            $round: [
                { $multiply: [{ $divide: ["$ultra_budget_count", "$listing_count"] }, 100] },
                1
            ]
        },
        pct_budget: {
            $round: [
                { $multiply: [{ $divide: ["$budget_count", "$listing_count"] }, 100] },
                1
            ]
        },
        pct_moderate: {
            $round: [
                { $multiply: [{ $divide: ["$moderate_count", "$listing_count"] }, 100] },
                1
            ]
        },
        pct_premium: {
            $round: [
                { $multiply: [{ $divide: ["$premium_count", "$listing_count"] }, 100] },
                1
            ]
        }
```

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```
        }
    },
    {
        $sort: { avg_total_per_night: 1 }
    }
])


db = db.getSiblingDB("sample_airbnb")
db.listingsAndReviews.aggregate([
    {
        $match: {
            "address.market": { $in: ["Maui", "Kauai", "The Big Island", "Oahu"] },
            accommodates: { $gte: 4 },
            bedrooms: { $gte: 2 }
        }
    },
    {
        $project: {
            market: "$address.market",
            price: { $toDouble: "$price" },
            cleaning_fee: {
                $cond: [
                    { $ifNull: ["$cleaning_fee", false] },
                    { $toDouble: "$cleaning_fee" },
                    0
                ]
            },
            accommodates: 1,
            bedrooms: 1,
            has_kitchen: { $in: ["Kitchen", "$amenities"] },
            has_washer: { $in: ["Washer", "$amenities"] },
            has_parking: {
                $or: [
                    { $in: ["Free parking on premises", "$amenities"] },
                    { $in: ["Free street parking", "$amenities"] }
                ]
            },
            per_night_total: {
```

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```
            $divide: [
                {
                    $add: [
                        { $multiply: [{ $toDouble: "$price" }, 3] },
                        {
                            $cond: [
                                { $ifNull: ["$cleaning_fee", false] },
                                { $toDouble: "$cleaning_fee" },
                                0
                            ]
                        }
                    ]
                },
                3
            ]
        },
        cost_per_person: {
            $divide: [
                {
                    $divide: [
                        {
                            $add: [
                                { $multiply: [{ $toDouble: "$price" }, 3] },
                                {
                                    $cond: [
                                        { $ifNull: ["$cleaning_fee", false] },
                                        { $toDouble: "$cleaning_fee" },
                                        0
                                    ]
                                }
                            ]
                        },
                        3
                    ]
                },
                "$accommodates"
            ]
        }
    }
```

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```
  },
  {
    $group: {
      _id: "$market",
      avg_nightly_price: { $avg: "$price" },
      avg_total_per_night: { $avg: "$per_night_total" },
      avg_cost_per_person: { $avg: "$cost_per_person" },
      listing_count: { $sum: 1 },
      pct_with_kitchen: {
        $avg: { $cond: ["$has_kitchen", 1, 0] }
      },
      pct_with_washer: {
        $avg: { $cond: ["$has_washer", 1, 0] }
      },
      pct_with_parking: {
        $avg: { $cond: ["$has_parking", 1, 0] }
      }
    }
  },
  {
    $project: {
      _id: 0,
      market: "$_id",
      avg_nightly_price: { $round: ["$avg_nightly_price", 2] },
      avg_total_per_night: { $round: ["$avg_total_per_night", 2] },
      avg_cost_per_person: { $round: ["$avg_cost_per_person", 2] },
      total_listings: "$listing_count",
      pct_with_kitchen: { $round: [{ $multiply: ["$pct_with_kitchen", 100] }, 0] },
      pct_with_washer: { $round: [{ $multiply: ["$pct_with_washer", 100] }, 0] },
      pct_with_parking: { $round: [{ $multiply: ["$pct_with_parking", 100] }, 0] }
    }
  },
  {
    $sort: { avg_cost_per_person: 1 }
  }
])
```

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Question 2:

```
db = db.getSiblingDB("nba")

// 1990s analysis
db.games.aggregate([
  {
    $match: {
      date: {
        $gte: ISODate("1990-01-01T00:00:00.000Z"),
        $lt: ISODate("2000-01-01T00:00:00.000Z")
      }
    }
  },
  {
    $unwind: "$box"
  },
  {
    $group: {
      _id: null,
      total_games: { $sum: 1 },
      total_fg3: { $sum: "$box.team.fg3" },
      total_fg3a: { $sum: "$box.team.fg3a" },
      total_fg: { $sum: "$box.team.fg" },
      total_fga: { $sum: "$box.team.fga" },
      total_pts: { $sum: "$box.team.pts" },
      avg_fg3_per_game: { $avg: "$box.team.fg3" },
      avg_fg3a_per_game: { $avg: "$box.team.fg3a" }
    }
  },
  {
    $project: {
      _id: 0,
      decade: "1990s",
      total_games: 1,
      avg_3pt_made_per_game: { $round: ["$avg_fg3_per_game", 2] },
      avg_3pt_attempts_per_game: { $round: ["$avg_fg3a_per_game", 2] },
      three_pt_pct: {
        $round: [
          { $multiply: [{ $divide: ["$total_fg3", "$total_fg3a"] }, 100] },
          1
        ]
      },
      three_pt_attempt_rate: {
        $round: [
          { $multiply: [{ $divide: ["$total_fg3a", "$total_fga"] }, 100] },
```

The Rise of the Three-Pointer: Offensive Evolution in the NBA (1990s vs. 2000s)

```
                1
            ]
        },
        pct_points_from_3pt: {
            $round: [
                { $multiply: [
                    { $divide: [
                        { $multiply: ["$total_fg3", 3] },
                        "$total_pts"
                    ]},
                    100
                ]},
                1
            ]
        }
    }
}
])

db = db.getSiblingDB("nba")

// 2000s analysis
db.games.aggregate([
    {
        $match: {
            date: {
                $gte: ISODate("2000-01-01T00:00:00.000Z"),
                $lt: ISODate("2010-01-01T00:00:00.000Z")
            }
        }
    },
    {
        $unwind: "$box"
    },
    {
        $group: {
            _id: null,
            total_games: { $sum: 1 },
            total_fg3: { $sum: "$box.team.fg3" },
            total_fg3a: { $sum: "$box.team.fg3a" },
            total_fg: { $sum: "$box.team.fg" },
            total_fga: { $sum: "$box.team.fga" },
            total_pts: { $sum: "$box.team.pts" },
            avg_fg3_per_game: { $avg: "$box.team.fg3" },
            avg_fg3a_per_game: { $avg: "$box.team.fg3a" }
```

The Rise of the Three-Pointer: Offensive Evolution in the NBA (1990s vs. 2000s)

```
      }
   },
   {
      $project: {
         _id: 0,
         decade: "2000s",
         total_games: 1,
         avg_3pt_made_per_game: { $round: ["$avg_fg3_per_game", 2] },
         avg_3pt_attempts_per_game: { $round: ["$avg_fg3a_per_game", 2] },
         three_pt_pct: {
            $round: [
               { $multiply: [{ $divide: ["$total_fg3", "$total_fg3a"] }, 100] },
               1
            ]
         },
         three_pt_attempt_rate: {
            $round: [
               { $multiply: [{ $divide: ["$total_fg3a", "$total_fga"] }, 100] },
               1
            ]
         },
         pct_points_from_3pt: {
            $round: [
               { $multiply: [
                  { $divide: [
                     { $multiply: ["$total_fg3", 3] },
                     "$total_pts"
                  ]},
                  100
               ]},
               1
            ]
         }
      }
   }
])

db = db.getSiblingDB("nba")

db.games.aggregate([
   {
      $match: {
         date: {
            $gte: ISODate("1990-01-01T00:00:00.000Z"),
            $lt: ISODate("2010-01-01T00:00:00.000Z")
```

The Rise of the Three-Pointer: Offensive Evolution in the NBA (1990s vs. 2000s)

```
        }
      }
    },
    {
      $project: {
        year: { $year: "$date" },
        box: 1
      }
    },
    {
      $unwind: "$box"
    },
    {
      $group: {
        _id: "$year",
        avg_3pt_made: { $avg: "$box.team.fg3" },
        avg_3pt_attempts: { $avg: "$box.team.fg3a" },
        total_fg3: { $sum: "$box.team.fg3" },
        total_fg3a: { $sum: "$box.team.fg3a" },
        total_fga: { $sum: "$box.team.fga" },
        total_pts: { $sum: "$box.team.pts" }
      }
    },
    {
      $project: {
        _id: 0,
        year: "$_id",
        avg_3pt_made: { $round: ["$avg_3pt_made", 2] },
        avg_3pt_attempts: { $round: ["$avg_3pt_attempts", 2] },
        three_pt_attempt_rate: {
          $round: [
            { $multiply: [{ $divide: ["$total_fg3a", "$total_fga"] }, 100] },
            1
          ]
        },
        pct_points_from_3pt: {
          $round: [
            { $multiply: [
              { $divide: [
                { $multiply: ["$total_fg3", 3] },
                "$total_pts"
              ]},
              100
            ]},
            1
```

The Rise of the Three-Pointer: Offensive Evolution in the NBA (1990s vs. 2000s)

```
            ]
        }
      }
    },
    {
      $sort: { year: 1 }
    }
])
```